OC Core 1.3.1 Methods or Simulation Companion EN

Document boundary

- This companion collects operational, empirical, proof, simulation, and dataset routes from the real corpus.
- It exists to make the reproducibility and simulation layer inspectable without improvising a second theory.
- Simulation and dataset surfaces stay bounded support layers and do not silently promote claims.

Contents

1 Cross-Disciplinary Instantiations of the Continuum Framework

This chapter tests a bounded representation claim: selected scientific and formal-theoretical domains can be expressed by the tuple

$$(\Omega, A, P(t), J(t), \Theta, \partial\Omega, C, k(t))$$

in ways consistent with the structural hierarchy of levels K_0–K_{12}, the threshold landscape, the operator family, and the branch topology of Section **??**. Here $k(t)$ suppresses the continuum index only to keep the domain tuple readable; locally it denotes the corresponding $k(K_x, t)$. Likewise, later domain blocks may suppress t in symbols such as P_x or J_x when the paragraph describes the family of potentials or flows rather than a single time slice. Each domain-specific mapping is treated here under one canonical label: the *candidate-embedding rule*. Phrases such as bounded representation test, structural-compatibility screen, bounded embedding pass, and bounded embedding criterion are local descriptions of that rule, not separate criteria. The aim is to show how well-studied classes of systems can be represented as continua under OC's structural axioms, not to claim that every object in a discipline is exhausted by a single OC encoding (**anderson_more_is_different**).

This is a structural-compatibility claim, not an operational-advantage claim. The practical comparison is deferred to Section **??** and Appendix **??**. This chapter asks whether a domain can be represented as a bounded continuum; it does not claim that every such domain already has a closed theorem packet, held-out replay, comparator audit, or deployment-ready procedure.

1.1 Principles for Domain Embedding

To represent a discipline within OC, the following structural alignment conditions must be satisfied.

1. Mapping of objects to continuum components. A domain has a candidate continuum representation if its objects can be mapped to:

- Ω: admissible configurations,
- A: axes defining essential degrees of freedom,
- P: potentials encoding forces, energies, costs, or semantic tension,
- J: flows (matter, charge, probability, information, resources),
- Θ: thresholds governing stability, phase change, and collapse,
- $\partial\Omega$: viability boundaries,
- C: cycles maintaining persistence,
- $k(t)$: continuumness (viability measure).

2. Embedding-space constraints. A valid instantiation requires both $\Omega(K_x) \subseteq \Omega(M_x)$ and $A(K_x) \subseteq A(M_x)$. If the required states or axes are absent from the embedding space, the proposed instantiation is structurally inadmissible.

3. K-level correspondence. A discipline is boundedly embedded at level K_x only when:

- its dominant axes correspond to the axes of K_x,
- its thresholds refine those of lower levels,
- its flows and cycles match the operator family of K_x,
- its behaviour respects dimensional monotonicity.

4. Structural compatibility. Domain instantiation requires:

- internal consistency with OC thresholds,
- existence of at least one supporting cycle C_{support},
- nonempty admissible region $\Omega \neq \emptyset$,
- extendibility of domain dynamics through the operator family.

A proposed embedding fails if any check fails: the mapping has an empty admissible region, the required axes are absent from the embedding space, the declared thresholds cannot be satisfied, or the proposed cycles do not keep the object away from the relevant boundary. These are rejection conditions, not merely cosmetic omissions.

The following sections summarise bounded candidate representations across disciplines.

1.2 Physics (Level K_2)

Physical systems instantiate K_2 when field configurations, order parameters, and coherence structures form the relevant axes. The discussion is intentionally bounded by standard critical phenomena, renormalisation, and percolation settings rather than by an unrestricted claim about all of physics (**goldenfeld_lectures**; **wilson_renormalization**; **stauffer_percolation**).

State space and axes.

- Ω_2: field configurations (scalar, vector, gauge fields).
- A_2: spatial axes, mode axes, internal symmetry axes, order-parameter axes.

Potentials and flows.

- P_2: physical energy functionals (action, free energy).
- J_2: diffusive, convective, and field-theoretic currents.

Thresholds.

- Θ_{crit}: phase boundaries,
- Θ_{BKT}: Berezinskii–Kosterlitz–Thouless vortex unbinding threshold,
- Θ_{mass}: coherence threshold for mass emergence,
- Θ_{death}: confinement or decoupling conditions.

Boundary structures. Within this bounded mapping, $\partial\Omega$ corresponds to critical surfaces, percolation boundaries, and phase-separation interfaces.

Cycles. Topological solitons, vortex pairs, renormalisation cycles, and oscillatory field modes are instances of C_2.

Bounded embedding criterion (Physics → K_2). A physical system meets the bounded K_2 embedding criterion when:

- its fields define a connected configuration space Ω_2,
- its energy functional defines thresholds compatible with OC,
- flows obey conservation laws expressible via J_2,
- coherent cycles maintain structural persistence.

1.3 Chemistry and Origins of Life (Levels K_3–K_4)

Chemical systems are represented at K_3, while protocellular systems are represented at K_4. These mappings are bounded by reaction-network closure, catalytic organisation, and membrane-constrained prebiotic structure (**kauffman_autocatalytic**; **raf_networks**).

Chemical continua (K_3).

- Ω_{chem}: concentration vectors, reaction states, catalytic configurations.
- A_{chem}: species concentrations, environmental parameters (pH, salinity, temperature).
- $P_{\text{chem}}(t)$: chemical potentials, affinity gradients, and concentration disequilibria.
- J_{chem}: reaction rates, transport fluxes.
- Θ_{act}: catalytic activation threshold.
- Θ_{closure}: reflexively autocatalytic and food-generated (RAF) closure threshold.
- C_{chem}: reaction cycles, RAF closure loops.

Protocellular continua (K_4).

- Ω_4: vesicle states, membrane configurations, internal chemical compositions.
- A_4: curvature, surface tension, permeability, charge.
- $P_4(t)$: osmotic, chemical, electrochemical, and curvature potentials.
- $\Theta_{\text{perm}}, \Theta_{\text{osm}}, \Theta_{\text{curv}}$: membrane viability thresholds.
- J_4: osmotic fluxes, ion fluxes, metabolic flows.
- C_4: metabolic cycles, membrane-maintenance cycles, gradient-sustaining cycles.

Bounded embedding criterion (Chemistry → K_3). A chemical system meets the bounded K_3 embedding criterion when:

- reaction networks form a nonempty admissible region,
- catalytic and closure thresholds are representable without membrane-specific axes,
- cycles maintain reaction closure and concentration control,
- embedding constraints support the chemical axes required by the route.

Bounded embedding criterion (Protocells → K_4). A protocellular system meets the bounded K_4 embedding criterion when:

- membrane-supported chemical dynamics form a nonempty admissible region,
- membrane boundaries define $\partial\Omega_4$,
- cycles maintain concentration gradients and compartment integrity,
- embedding constraints support both chemical and membrane axes.

1.4 Biology (Level K_5)

Biological cells, beyond the protocellular K_4 route, emerge when protocells acquire excitability and structured regulation. Here the target is the bounded embedding of excitable and morphogenetic organisation rather than an unrestricted theory of life as such (**turing_morphogenesis**; **hopfield_nn**).

State space and axes.

- Ω_5: distributions of membrane potential, channel states, ion gradients.
- A_5: excitability axes (membrane potential ΔV, ion concentrations, gating-variable axes).

Potentials and flows.

$$P_5 : \text{electrochemical potentials}, \qquad J_5 : \text{ion currents, metabolic fluxes}.$$

Thresholds.

$$\Theta_{\text{exc}}, \quad \Theta_{\text{refr}}, \quad \Theta_{\text{grad}}, \quad \Theta_{\text{death}}.$$

Cycles. The relevant cycles include the proto-spike cycle C_{spike}, metabolic cycles, and electrical oscillation loops.

Bounded embedding criterion (Biology → K_5). A biological system meets the bounded K_5 embedding criterion when:

- excitability defines nontrivial ΔV-axes,
- thresholds define firing and refractory structure,
- cycles maintain homeostasis and signalling,
- embedding includes ionic, energetic, and membrane-support axes.

1.5 Cognition (Level K_6)

The K_6–K_{10} sections below are structural candidate embeddings. They are frontier-facing representation sketches unless Section **??** or Appendix **??** explicitly promotes a narrower operational lane. They are not usable-now practical claims by themselves.

Cognitive systems arise when representations, bindings, and predictive models become the dominant axes. The discussion is anchored to classical work on the organisation of behaviour and predictive-processing motifs, not to a claim that OC replaces cognitive science in full (**hebb_organization**; **friston_free_energy**).

State space and axes.

- Ω_6: representational configurations, activated conceptual states, prediction states.
- A_6: representational axes, binding axes, predictive axes.

Potentials and flows.

- P_6: prediction error, semantic tension, confidence.
- J_6: attention flows, inference flows, memory transitions.

Thresholds.

$$\Theta_{\text{pred}}, \quad \Theta_{\text{bind}}, \quad \Theta_{\text{coh}}, \quad \Theta_{\text{expr}}.$$

Cycles. The relevant cycles include attention-prediction cycles, memory-consolidation cycles, and model-stability loops.

Bounded embedding criterion (Cognition → K_6). A cognitive system meets the bounded K_6 embedding criterion when:

- it maintains coherent internal models within expressivity thresholds,
- flows reduce prediction error and sustain conceptual structure,
- cycles regulate attention, memory, and updating,
- boundaries $\partial\Omega_6$ encode model limits.

1.6 Society (Level K_7)

Social systems instantiate K_7 when norms, institutions, and trust form the dominant structural axes. The intended scope is thresholded collective behaviour, networked coordination, and institutional persistence (**granovetter_threshold**; **schelling_micromotives**; **newman_networks**).

State space and axes.

- Ω_7: communication graphs, role structures, institutional states.
- A_7: normative axes, trust axes, institutional axes.

Potentials and flows.

$$P_7 : \text{trust, legitimacy, symbolic capital,}$$
$$J_7 : \text{communication, resource exchange, and norm transmission.}$$

Thresholds.

$$\Theta_{\text{trust}}, \quad \Theta_{\text{leg}}, \quad \Theta_{\text{role}}.$$

Cycles. The relevant cycles include institutional cycles, normative cycles, and coordination cycles.

Bounded embedding criterion (Society → K_7). A social system meets the bounded K_7 embedding criterion when:

- trust and norms define viability thresholds,
- institutions maintain cycles of governance and coordination,
- flows of communication and resources respect embedding constraints,
- social boundaries form nonempty $\partial\Omega_7$.

1.7 Civilizational Systems (Level K_8)

Civilizational systems extend social continua to encompass infrastructures, energy flows, and large-scale coordination. The scope is limited to large-scale organised systems with networked infrastructure and stability constraints (**newman_networks**; **may_stability_complexity**).

State space and axes.

- Ω_8: infrastructure networks, economic states, population distributions.
- A_8: infrastructural axes, energy-flow axes, communication axes, complexity axes.

Potentials and flows.

$$P_8 : \text{energy availability, resource gradients, risk potentials,} \qquad J_8 : \text{trade, energy flux, data flows.}$$

Thresholds.

$$\Theta_{\text{stress}}, \quad \Theta_{\text{capacity}}, \quad \Theta_{\text{cohesion}}.$$

Cycles. The relevant cycles include economic cycles, infrastructure-renewal cycles, and knowledge-transfer loops.

Bounded embedding criterion (Civilizational systems → K_8). A civilizational system meets the bounded K_8 embedding criterion when:

- energy and resource structures define P_8,
- infrastructural flows form stable global cycles,
- institutions at K_7 embed consistently as substructures,
- large-scale boundaries define global $\partial\Omega_8$.

1.8 Theory and Formal Systems (Levels K_9–K_{10})

Theory-level and metatheoretical formal-system structures occupy the upper route treated in this chapter. This subsection gives an OC-internal schema for K_9–K_{10}, with local support carried by Sections **??** and **??**. It does not claim a fully closed external embedding theorem for every theory or formal system. Recursive structural continua remain the K_{11} layer outside this chapter's candidate-embedding rule.

Theoretical continua (K_9).

- Ω_9: space of models, formalisms, and representations.
- A_9: conceptual, formal, and modelling axes.
- P_9: coherence potentials, empirical tension, explanatory power.
- J_9: model-revision flows, anomaly-handling flows, and argument-transfer flows.
- $\Theta_{\text{expr}}, \Theta_{\text{coh}}$: thresholds for consistency and expressivity.
- $\partial\Omega_9$: inconsistency, expressivity-failure, or unrepaired-anomaly boundaries.
- C_9: paradigm cycles, research cycles.
- $k_9(t)$: coherence and validation-bandwidth continuumness.

Metatheoretical formal-system continua (K_{10}).

- Ω_{10}: space of metatheoretical states, formal systems, proof states, model-of-models objects, and recursion-bounded languages.
- A_{10}: metatheory axes, recursion axes, consistency axes, expressivity axes, and theory-comparison axes.
- P_{10}: proof-flow capacity, consistency margin, formal expressive capacity, and metatheoretical coherence potential.
- J_{10}: proof-search flows, formal-update flows, meta-model translations, and recursion-control flows.
- $\Theta_{\text{cons}}, \Theta_{\text{rec}}, \Theta_{\text{meta}}$: thresholds for consistency, bounded recursion, and meta-level coherence.
- $\partial\Omega_{10}$: inconsistency, undecidable-overreach, recursion-boundary surfaces, or meta-theory boundary failures.
- C_{10}: proof cycles, recursion cycles, formal-update cycles, and model-of-models audit cycles.

- $k_{10}(t)$: formal and metatheoretical viability under consistency, recursion, self-reference, and proof-throughput constraints.

Candidate-embedding rule (Theory → K_9, metatheoretical formal system → K_{10}). The theoretical and metatheoretical formal-system routes meet the candidate-embedding rule when the relevant K_9 or K_{10} structure satisfies the following conditions:

- its conceptual and formal axes form a valid axis set,
- its coherence constraints define thresholds compatible with OC,
- cycles of reasoning, modelling, and revision sustain viability,
- embedding spaces support the required expressive degrees of freedom.

Summary

This chapter states bounded cross-disciplinary candidate-embedding criteria across physics, chemistry, protocells, biology, cognition, society, civilizational systems, theory, and formal systems. Each bounded route is mapped to the same structural tuple at the level of detail declared in its subsection, but each mapping is claimed only at the level of structural compatibility, not as a sweeping proof that OC already replaces the best local models in every discipline. That stronger operational and comparative question is taken up later under the explicit support classes in Section **??** and the row-level atlas in Appendix **??**.

2 Extended Falsifiability and Experimental Programme

This section restores and expands the falsifiability framework for the Ontology of Continua (OC). It formulates explicit structural predictions, falsification criteria, and experimental programmes across the hierarchy K_2–K_{12}, using only the axioms and definitions of Core 1.3.

The goal is not to provide a full catalogue of experiments but to show that OC is structurally falsifiable: it makes cross-domain commitments about thresholds, cycles, dimensional transitions and boundary behaviour that can in principle be contradicted by empirical and theoretical data.

2.1 Meaning of Falsifiability for Structural Theories

For a structural theory, falsifiability is not limited to individual numerical predictions. Instead, falsification occurs when the structural relations between

$$(\Omega, A, P(t), J(t), \Theta, \partial\Omega, C, k(t))$$

are violated in ways incompatible with the axioms and theorems of OC.

Structural vs phenomenological falsifiability.

- *Phenomenological falsifiability* concerns direct mismatches between model outputs and observations (e.g. wrong phase diagram, incorrect scaling exponent).
- *Structural falsifiability* concerns violations of the universal constraints on continua: thresholds, cycles, dimensional monotonicity, embedding-compatibility, collapse and rebirth conditions.

OC primarily lives at the structural level; phenomenological models must factor through OC's structural requirements to be considered proper instantiations.

Structural falsification channels. A continuum model claiming to instantiate OC can be falsified if:

1. **Threshold violations** occur systematically: observed systems remain viable while violating putative Θ_{exist}, Θ_{stab}, Θ_{crit} or Θ_{death}.
2. **Cycle disappearance** contradicts viability: systems maintain long-term organisation without any identifiable cycle complex C that satisfies the OC conditions.
3. **Forbidden dimensional transitions** are observed: dimension appears to increase without new axes in the embedding space or without crossing a dimensional threshold Θ_{dim}.
4. **Inconsistent tuples** are required: successful models require mutually incompatible versions of the canonical OC state tuple for one and the same system.

Structural vs empirical failure. Empirical failure of a specific domain model does not immediately falsify OC; the structural theory is challenged only when the empirical success of alternative models requires violations of OC's structural constraints. Conversely, if a domain repeatedly exhibits structures that OC forbids (e.g. stable continua without supporting cycles), OC is directly falsified.

Cross-domain falsification logic. Because OC claims universality, a structural falsification in one domain (propely established) propagates across all domains that rely on the same axioms. This makes OC *more* falsifiable than narrow theories:

- if monotonicity of dimension fails in a controlled physical system, it must also be reconsidered for cognitive and social continua;
- if viable protocells are demonstrated without any boundary structure compatible with $\partial\Omega$, the entire K-level hierarchy is affected.

The remainder of this chapter formulates explicit predictions and tests.

2.2 Predictions and Tests in Physics (K_2)

Physical continua provide the most classical testing ground for OC at level K_2.

Structural predictions.

- **P2.1 (Critical thresholds).** Phase transitions correspond to structural crossings of Θ_{crit}. No sharp long-range ordering transition may occur without a corresponding threshold surface in Ω_2.
- **P2.2 (Percolation monotonicity).** For connectivity-based systems with control parameter p, global viability ($k_2 > 0$) requires $p \geq p_c$; for $p < p_c$ no infinite cluster compatible with Ω_2 can exist.
- **P2.3 (Dimensional monotonicity).** No physical continuum can increase its effective dimension without the activation of at least one new axis in the embedding space M_2 and crossing a dimensional threshold Θ_{dim}.
- **P2.4 (Boundary-driven collapse).** Collapse of ordered phases occurs via destruction of supporting cycles (e.g. vortex pairs, domain structures) and loss of well-defined $\partial\Omega_2$, not via spontaneous disappearance of order in the interior alone.

Falsification criteria and tests.

- **F2.1.** A robust, reproducible phase transition with no structural thresholds in the order-parameter geometry would falsify P2.1.
- **F2.2.** Observation of stable infinite clusters for $p < p_c$ across well-controlled percolation-like systems would contradict P2.2.

- **F2.3.** Demonstration of a continuum whose effective dimensionality increases (e.g. from 2D to 3D behaviour) while all axes of M_2 remain fixed and no new degrees of freedom are activated would falsify P2.3 and the general dimensional monotonicity theorem.
- **Experimental programme.** Precision experiments on coherence destruction, vortex unbinding, and percolation thresholds can be used to map empirical Θ_{crit} and test whether the associated $\partial\Omega_2$ behaves as OC predicts (critical slowing down, boundary localisation of collapse).

2.3 Predictions and Tests in Origins of Life (K_3–K_4)

For origins-of-life scenarios OC makes explicit claims about catalytic closure, membrane structure and protocell viability.

Structural predictions.

- **P3.1 (Catalytic closure).** Long-term stable chemical organisations at K_3 require at least one RAF-like cycle complex; systems without catalytic closure cannot maintain $k_3 > 0$.
- **P3.2 (Osmotic collapse threshold).** Protocells with fixed membrane composition exhibit a radius–tension– permeability relation: beyond a critical combination of osmotic gradient, curvature and permeability ($\Theta_{\text{grad}}, \Theta_{\text{perm}}, \Theta_{\text{mem}}$) collapse is inevitable.
- **P3.3 (Boundary necessity).** Protocells lacking a structurally identifiable boundary $\partial\Omega_4$ (membrane or equivalent compartment structure) cannot sustain nontrivial metabolic flows $J_{\text{metabolic}}$ over timescales defining $k_4 > 0$.
- **P3.4 (Patch-localised failure).** Membrane collapse initiates on boundary patches where local thresholds are first crossed, not uniformly across the surface.

Falsification criteria and tests.

- **F3.1.** Demonstration of chemically self-sustaining organisations with $k_3 > 0$ and no identifiable catalytic closure or cycle complex would falsify P3.1.
- **F3.2.** Systematic violation of predicted osmotic-collapse relations (e.g. arbitrary gradients sustained without structural adjustment) would contradict P3.2.
- **F3.3.** Observation of long-lived protocell-like systems with robust metabolic fluxes but no boundary structure compatible with $\partial\Omega_4$ would falsify P3.3 and weaken the role of boundaries in OC.
- **Experimental programme.** Controlled protocell leakage and bursting experiments, combined with quantitative measurements of gradients and membrane properties, can test the structural status of Θ_{grad}, Θ_{perm}, Θ_{mem} and patch dynamics.

2.4 Predictions and Tests in Biological Systems (K_4–K_5)

At the biological level, OC focuses on excitability, proto-spikes and the transition $K_4 \to K_5$.

Structural predictions.

- **P4.1 (Minimal excitability threshold).** The transition from purely metabolic protocells to excitable continua requires a minimal excitability threshold Θ_{exc}; no stable K_5 behaviour can exist without nontrivial ΔV-axes and a well-defined spike threshold.
- **P4.2 (Proto-spike invariants).** Across early excitable systems there exist minimal invariants for spike-like cycles (e.g. minimal amplitude of ΔV, refractory period τ) corresponding to structural constraints of C_{spike}.

- **P4.3 (Excitability precedes complex inheritance).** Long-term inheritance of complex regulatory structures requires prior establishment of K_5-like channels and excitability cycles; purely K_4 systems cannot indefinitely sustain inherited regulatory complexity without transitioning to a higher level.

Falsification criteria and tests.

- **F4.1.** Demonstration of genuinely K_5-like signalling (spike trains, excitable waves) in systems with no identifiable excitability axes or thresholds would falsify P4.1.
- **F4.2.** Discovery of stable, complex inheritance mechanisms in protocell systems strictly confined to K_4 (no excitability, no spike-like cycles) would challenge P4.3.
- **Experimental programme.** Minimal ion-channel sets, synthetic excitable membranes and controlled ΔV instability tests can be used to map the structural space of Θ_{exc}, Θ_{refr} and the resulting C_{spike}.

2.5 Predictions and Tests in Cognitive Continua (K_6)

For cognitive systems OC emphasises binding capacity, prediction error and cycle-based maintenance of internal models.

Structural predictions.

- **P6.1 (Binding capacity limit).** For any cognitive continuum there is a finite binding capacity threshold Θ_{bind} beyond which additional features cannot be integrated into coherent representations without collapse of k_6.
- **P6.2 (Prediction-error collapse).** There exists a prediction-error threshold Θ_{pred} beyond which no coherent internal model can be maintained; cognitive collapse occurs when prediction error cannot be reduced by any flows J_6 within current axes.
- **P6.3 (Cycle-based collapse).** Cognitive collapse always involves destruction of key cycles (attention, prediction, memory); there is no collapse scenario with $C_{\max}(K_6)$ intact.

Falsification criteria and tests.

- **F6.1.** Robust evidence for unbounded binding capacity in finite systems (no observable thresholds, no degradation) would falsify P6.1.
- **F6.2.** Empirical regimes where prediction error diverges yet stable, coherent internal models persist indefinitely would contradict P6.2.
- **F6.3.** Demonstration of cognitive collapse in which attention, prediction and memory cycles remain fully intact would falsify P6.3 and the cycle-based definition of collapse.
- **Experimental programme.** Behavioural and neurophysiological studies of binding degradation under load, prediction-error saturation, and memory reconsolidation can probe the empirical counterparts of Θ_{bind}, Θ_{pred}, and C_{memory}.

2.6 Predictions and Tests in Social/Civilisational Continua (K_7–K_8)

At higher social levels OC proposes structural constraints on trust, institutional cycles and infrastructural stability.

Structural predictions.

- **P7.1 (Trust threshold).** Stable social continua at K_7 require trust above a threshold Θ_{trust}; below this threshold institutional cycles cannot maintain $k_7 > 0$.

- **P7.2 (Institutional cycle closure).** No stable K_7 continuum exists without at least one closed institutional cycle (e.g. taxation–service–legitimacy loop) compatible with OC's definition of C_7.
- **P8.1 (Infrastructure cycle thresholds).** Civilisational continua K_8 require stable infrastructural cycles; collapse at K_8 involves boundary failure in $\partial\Omega_8$ and percolation of failure across key infrastructural layers.

Falsification criteria and tests.

- **F7.1.** Historical or simulated examples of long-term stable institutions with trust consistently below any plausible Θ_{trust} would challenge P7.1.
- **F7.2.** Demonstration of durable, high-complexity societies without any identifiable closed governance or resource cycles would falsify P7.2.
- **F8.1.** Robust models of civilisational collapse that do not involve breakdown of infrastructural cycles or boundary failures in $\partial\Omega_8$ would contradict P8.1.
- **Experimental programme.** Quantitative analyses of historical data, network fragility models, and large-scale simulations of trust dynamics and infrastructure percolation can be used to test the structural role of Θ_{trust}, C_{inst} and $\partial\Omega_8$.

2.7 Theoretical and Meta-Theoretical Predictions (K_9–K_{10})

OC itself, and other high-level frameworks, can be tested at the theoretical and meta-theoretical levels.

Structural predictions.

- **P9.1 (Coherence thresholds).** Any long-lived theoretical continuum K_9 must satisfy coherence and expressivity thresholds $\Theta_{\text{coh}}, \Theta_{\text{expr}}$; persistent success of theories that explicitly violate internal coherence would contradict the OC framework.
- **P9.2 (Inconsistency cascades).** Accumulated inconsistencies in a theoretical framework must lead to collapse of its cycle complex (research programmes, paradigm loops); sustained progress in the presence of arbitrarily large incoherence is incompatible with OC.
- **P10.1 (Meta-incompleteness).** Any meta-theoretical continuum K_{10} is structurally incomplete: as bodies of models grow, structural tension at K_{10} must eventually exceed some Θ_{dim}, forcing either expansion of axes or collapse.

Falsification criteria and tests.

- **F9.1.** A fully self-consistent, closed theoretical system that captures all relevant models while never requiring expansion or revision and yet continues to support unbounded empirical progress would challenge P9.2 and P10.1.
- **F9.2.** Demonstration of theoretical frameworks that remain incoherent by OC standards but nonetheless exhibit stable, cumulative, cross-domain predictive success would falsify the structural link between coherence and k_9.
- **Experimental programme.** Formal analyses of model families, logical consistency checks, and meta-level studies of theory change can be used to test the existence and behaviour of Θ_{expr}, Θ_{coh} and associated cycles at K_9–K_{10}.

2.8 Meta-Criteria and Cross-Domain Validation

Finally, OC proposes meta-level criteria that couple falsifiability across domains and K-levels.

Predictive invariants across K-levels. Several structural relations are predicted to hold uniformly:

- existence of at least one supporting cycle C_{support} for any live continuum;
- monotonicity of dimension under birth operators;
- necessity of embedding-space compatibility $A(K_x) \subseteq A(M_x)$ and threshold compatibility with M_x;
- collapse signatures: critical slowing, boundary localisation, cycle breakdown, divergence of structural tension.

Violation of these invariants at any well-understood K-level would propagate upward and downward in the hierarchy.

Algorithmic structural tests. Core 1.2 and subsequent work are expected to develop algorithmic tests for:

- Ω-consistency (nonempty admissible region given thresholds),
- cycle viability (existence of cycle complexes C under given flows),
- embedding compatibility (axes and thresholds vs. M_x).

These tests can be applied to concrete models in physics, chemistry, biology, cognition and social sciences to check whether they are legitimate OC instantiations.

Cross-domain falsification logic. Because axioms are shared across K-levels, falsification has a *cross-domain* structure:

- if one structural axiom fails in a tightly controlled physical context, all higher-level predictions derived from it must be reconsidered;
- conversely, consistent validation of the same structural motif (e.g. boundary-driven collapse) in multiple domains increases confidence in the corresponding OC axiom.

Unified validation pipeline. Core 1.3 provides the structural basis for an explicit validation pipeline in Core 1.2:

1. select domain models and identify their continuum components $(\Omega, A, P, J, \Theta, \partial\Omega, C, k)$;
2. test local structural predictions (thresholds, cycles, boundaries);
3. test cross-level invariants (dimensional monotonicity, embedding compatibility);
4. update the mapping between domain theories and K-levels.

OC is therefore not a purely conceptual framework: it is designed to be empirically constrained and — in principle — structurally falsifiable across the full hierarchy K_0–K_{10}.

3 Operationalization Program and Predictive Promotion Discipline

This chapter turns the theorem-bearing core into lawful operational discipline. Its job is to state how a packet becomes promotable, how bridge evidence and claim scope are bounded, and where operational language must stop. Comparator verdicts are summarized here only through promotion discipline and are handled explicitly in Chapter **??** and Appendix **??**. This chapter does not carry the empirical replay burden or the proof-machinery burden in full; those tasks are handled by Chapter 4 and Chapter 5.

This chapter is projected from the canonical science SPOT surface `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json` (surface id `OC_CORE_1_3_SCIENCE_SPOT`). The generator and validator make the alignment auditable: operational status, bridge evidence, and closure verdicts are checked against the same source-owned science corpus rather than edited as independent prose claims. The generated section below prints the exact metadata.repo_sha and metadata.ts_utc for the SPOT revision used in this build.

SPOT revision: repo SHA 0dd049c48def00df8a7d50bef07abb127a3c4fb3; timestamp 2026-04-22T09:23:17Z.

3.1 Why the present release separates explanatory force from predictive promotion

The canonical science SPOT distinguishes explanatory reach from lawful promotion. Empirical and predictive outward claims are promoted only when a domain closes one continuous trace from the kernel and K-level theorem spine through observables to held-out evidence and explicit falsifiers; theorem-native mathematical claims close by proof route rather than by measurement surface.

3.2 Current release truth

The canonical SPOT checks 5 core lanes. At present 5 lanes satisfy the full promotion bar, while 0 lanes remain fail-closed. That PASS count applies only to the five core theorem-native lanes inside CORE_1_3_SCIENCE_ONLY; frame-only, extension, and refuted rows remain support, frontier, or negative-audit rows rather than public positive closure claims. Mathematics is the formal anchor for exact proof obligations, while physics, chemistry, biology, and Systems / Civilizational projection now clear theorem-native closure on their locked held-out routes. Within the declared CORE_1_3_SCIENCE_ONLY scope, the hostile-review backlog is closed and the Core 1.3 science verdict is PASS.

3.3 Closed canon and frontier workbench

The program therefore runs in two layers. Closed canon admits theorem-native results only. Frontier workbench holds broader unified-science campaigns under explicit promotion gates and may not be advertised as closed science. The closed canon currently tracks 22 entities, while the frontier workbench tracks 2 active campaigns. Execution order remains DOMAIN_PRIORITY_BOARD_ONLY, and Phase 1 is tracked as a first-class dossier surface rather than as drifting editorial placeholder text.

3.4 Closure economics and anti-cycle law

Priority is ranked by (closure impact x dependency centrality x expected information gain) / (compute cost + data cost + measurement cost + editorial cost). Anti-cycle rule: A method class may not be repeated unless a new formal premise, new dataset family, or new observable family enters the claim.

3.5 Program board

Domain	Formal order	Cost order	Gain per cost	Phase	Launch rule
Mathematics	0	0	116.666667	PHASE_1_STABILIZE_CLOSED_CORE	MAINTAIN_ALWAYS_ON
Physics	1	1	64.8	PHASE_3_PHYSICS_CLOSURE	START_IMMEDIATELY_AFTER_PHASE_1
Chemistry	2	2	32.0	PHASE_4_CHEMISTRY_CLOSURE	START_AFTER_PHYSICS_THEOREM_PACKET_LOCKS
Biology	3	4	17.818182	PHASE_5_BIOLOGY_CLOSURE	START_AFTER_CHEMISTRY_THEOREM_PACKET_LOCKS

Domain	Formal order	Cost order	Gain per cost	Phase	Launch rule
Systems / Civilizational projection	4	3	`16.615385`	`PHASE_6_SYSTEMS_CLOSURE`	`ALLOW_PARALLEL_DATA_PREP_AFTER_CHEMISTRY_BUT_FINAL_PROMOTION_AFTER_BIOLOGY_FORMAL_ROUTE`

3.6 Domain order and promotion rule

The formal derivation order remains physics, chemistry, biology, and then Systems / Civilizational projection. The empirical cost order lets cheaper systems data preparation run early, but final systems promotion still waits for the biology formal route to lock. Bridge evidence may support the research program, but it cannot by itself promote a domain to closed science.

3.7 Operationalization table

Domain	Class	Trace	Evidence	Verdict	Next lawful action
Mathematics	`THEOREM_NATIVE`	`TRACE_COMPLETE`	`EVIDENCE_COMPLETE`	`PASS`	`MAINTAIN_REPLAY_DISCIPLINE`
Physics	`THEOREM_NATIVE`	`TRACE_COMPLETE`	`EVIDENCE_COMPLETE`	`PASS`	`MAINTENANCE_ONLY`
Chemistry	`THEOREM_NATIVE`	`TRACE_COMPLETE`	`EVIDENCE_COMPLETE`	`PASS`	`MAINTENANCE_ONLY`
Biology	`THEOREM_NATIVE`	`TRACE_COMPLETE`	`EVIDENCE_COMPLETE`	`PASS`	`MAINTENANCE_ONLY`
Systems / Civilizational projection	`THEOREM_NATIVE`	`TRACE_COMPLETE`	`EVIDENCE_COMPLETE`	`PASS`	`MAINTENANCE_ONLY`

3.8 Per-domain status

Mathematics. Wave `WAVE_0_ANCHOR` currently places the domain in class `THEOREM_NATIVE`, with trace status `TRACE_COMPLETE`, evidence status `EVIDENCE_COMPLETE`, and verdict `PASS`. Benchmark families cover `TIER1_AIMATH_PDF_013_Q010`, `strictclosureresultledger`, `counterexamplesearchcorpus`, `deterministicreplaytraces`. Measured outputs track `epsilon_star`, `minimal_denominator`, `minimal_augmentation_rank`, `component_fiber_count`, `move_connectivity_or_degree_bound`, `strict_terminalized_total`, `counterexample_executed_total`, `python_evidence_executed_total`. Measurement schema: Replay declared mathematical packets deterministically against the strict closure ledger and counterexample traces. This formal lane does not introduce a separate external theorem-to-observable route; it closes as a theorem-native mathematical anchor without an additional measurement surface. Acceptance criterion: No replay divergence and no surviving counterexample against the promoted packet. Falsifier: A single replay divergence or surviving counterexample collapses promotion for the affected packet. Primary route institutions: Clay Mathematics Institute, American Institute of Mathematics. Numerical packet / replay status: `PASS_ACTIVE_NUMERICAL_PACKET` / `PASS_REPLAYABLE`. Open gaps: none. Next lawful action: `MAINTAIN_REPLAY_DISCIPLINE`.

Physics. Wave `WAVE_1_PHYSICS` currently places the domain in class `THEOREM_NATIVE`, with trace status `TRACE_COMPLETE`, evidence status `EVIDENCE_COMPLETE`, and verdict `PASS`. Benchmark families cover `PHY_BENCH_001`, `fundamentalconstantsresiduals`, `spectrallineresidualenvelopes`, `transportorplasmascalingenvelopes`. Measured outputs track `relative_constant_error`, `sp ectral_line_residual, transport_scaling_residual. Measurement schema: Bind theorem-to-observable maps to pinned official constants and spectra, then evaluate held-out residual envelopes. The theorem-to-observable route has three claims. K1/K2 contradiction-load and stabilization-margin operators lawfully generate the bounded physics packet over constants, Balmer spectra, and transport-scaling families. Root-law thresholds are exposed as theorem-native residual observables on the pinned official routes rather than as unrestricted physical universalization claims. Every promoted benchmark family is traced through the locked K1/K2/K9/K10 route, official datasets, and held-out replay. Acceptance criterion: Held-out residuals stay inside the declared tolerance band for every promoted benchmark family. Falsifier: Any benchmark family with residuals outside tolerance or with broken sign/order constraints falsifies the promoted packet. Primary route institutions: NIST, U.S. Department of Energy Office of Science. Numerical packet / replay status: PASS_ACTIVE_NUMERICAL_PACKET / PASS_REPLAYABLE. Open gaps: none. Next lawful action: MAINTENANCE_ONLY.

Chemistry. Wave WAVE_2_CHEMISTRY currently places the domain in class THEOREM_NATIVE, with trace status TRACE_COMPLETE, evidence status EVIDENCE_COMPLETE, and verdict PASS. Benchmark families cover CHEM_BENCH_001, thermochemicalreferencevalues, spectroscopictransitionfamilies, kineticorequilibriumobservableenvelopes. Measured outputs track enthalpy_residual, spectral_transition_residual, kinetic_or_equilibrium_residual. Measurement schema: Map theorem packets to thermochemical, spectral, and kinetic observables against pinned reference tables. The theorem-to-observable route has three claims. K2/K3/K4 organizational thresholds lawfully generate the bounded chemistry packet over thermochemical, spectral, and kinetic observable families. The promoted chemistry packet binds source-native thresholds to pinned residual observables without expanding to unrestricted chemical universalization. Every promoted chemistry benchmark family is traced through the locked K2/K3/K4 route, pinned reference tables, and held-out replay. Acceptance criterion: Declared chemistry benchmark families pass the tolerance envelope on held-out compounds or reactions. Falsifier: A held-out thermochemical, spectral, or kinetic family outside tolerance collapses the promoted chemistry packet. Primary route institutions: NIST, U.S. Department of Energy Office of Science. Numerical packet / replay status: PASS_ACTIVE_NUMERICAL_PACKET / PASS_REPLAYABLE. Open gaps: none. Next lawful action: MAINTENANCE_ONLY.

Biology. Wave WAVE_3_BIOLOGY currently places the domain in class THEOREM_NATIVE, with trace status TRACE_COMPLETE, evidence status EVIDENCE_COMPLETE, and verdict PASS. Benchmark families cover BIO_BENCH_001, gene-expressiontime-seriesfamilies, cell-statetransitionsignatures, stimulus-responseorexcitabilityonsetmarkers. Measured outputs track expression_signature_residual, state_transition_order_error, response_onset_residual. Measurement schema: Choose one bounded biological lane and evaluate predicted state transitions or response signatures against open assay data. The theorem-to-observable route has three claims. K4/K5/K6/K7 organizational thresholds lawfully generate the bounded biology packet over expression, state-transition, and response-onset observable families. The promoted biology packet binds source-native biological organization claims to pinned assay observables without expanding to unrestricted biological universalization. Every promoted biological benchmark family is traced through the locked K4/K5/K6/K7 route, open assay datasets, and held-out replay. Acceptance criterion: The promoted biological packet reproduces the declared measurable signatures on held-out datasets without widening the claim boundary. Falsifier: If the bounded signature family fails on held-out datasets, the biological packet remains frontier-bound. Primary route institutions: NCBI GEO. Numerical packet / replay status: PASS_ACTIVE_NUMERICAL_PACKET / PASS_REPLAYABLE. Open gaps: none. Next lawful action: MAINTENANCE_ONLY.

Systems / Civilizational projection. Wave WAVE_4_SYSTEMS currently places the domain in class THEOREM_NATIVE, with trace status TRACE_COMPLETE, evidence status EVIDENCE_COMPLETE, and verdict PASS. Benchmark families cover SYS_BENCH_001, macroindicatortrajectories, resource-intensityorthroughputcurves, held-outintervalstabilityandregimeshifts. Measured outputs track trajectory_residual, turning_point_error, held_out_interval_regime_score. Measurement schema: Evaluate K8 projection packets against pinned official macro-process series with held-out interval tests. The theorem-to-observable route has three claims. K8 projection claims lawfully generate the bounded systems packet over measurable macro-process trajectories and regime-shift signatures on pinned official series. The promoted systems packet binds source-native K8 projection structure to residual and turning-point observables without authorizing unrestricted civilizational prediction. Every promoted systems benchmark family is traced through the locked K8 route, official macro-process data, and held-out interval replay. Acceptance criterion: Declared trajectory and regime-change observables remain inside tolerance on held-out intervals. Falsifier: A missed turning point or out-of-band held-out trajectory falsifies the promoted systems packet. Primary route institutions: World Bank, U.S. Department of Energy Office of Science. Numerical packet / replay status: PASS_

ACTIVE_NUMERICAL_PACKET / PASS_REPLAYABLE. Open gaps: none. Next lawful action: MAINTENANCE_ONLY.

3.9 Release consequence

The scientific SPOT therefore sets the Core 1.3 science verdict, within scope CORE_1_3_SCIENCE_ONLY, to PASS. The kernel, theorem spine, domain packets, held-out evidence, hostile-review dossiers, and unified atlas are now mutually aligned under the declared promotion bar.

Transition to empirical execution. Operational status alone is not enough. The next chapter fixes the evidence bar, held-out discipline, and replay conditions by which promoted packets stay scientifically lawful.

4 Hybrid-Escalation Evidence Bar and Execution Protocols

This chapter presents the empirical execution bar of the release. It states which source authorities, held-out designs, benchmark policies, tolerances, and falsifier conditions lawfully test promoted packets, and it separates bounded empirical support from rhetorical overreach. It does not restate the theorem spine, the full proof-support ledger, or the complete dataset inventory; it tells the reader how the packets are actually challenged.

The scientific authority for that evidence bar remains the canonical science SPOT, OC_CORE_1_3_SCIENCE_SPOT, materialised in this repository at releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json. This chapter projects that authority into reader-facing form so that the execution bar, replay discipline, and promotion verdict share one lawful source boundary. The generated section below prints the exact metadata.repo_sha and metadata.ts_utc for the SPOT revision used in this build.

SPOT revision: repo SHA 0dd049c48def00df8a7d50bef07abb127a3c4fb3; timestamp 2026-04-22T09:23:17Z. The evidence bar is now locked to the canonical SPOT rather than to drifting summary surfaces. The mathematics anchor is replay-only: it requires deterministic exact replay rather than a data-backed benchmark route. The empirical domain lanes require held-out results that clear the declared five-sigma bar without tail-breach inflation. Residual containment by itself is not enough for promotion.

4.1 Global evidence bar

For the empirical lanes, the default evidence bar remains hybrid escalation: official or open primary data first, then institute-run measurements only if the observable family cannot be closed otherwise. The mathematics lane is replay-only and closes through deterministic exact replay rather than an out-of-sample data route. In every lane, evidence discipline sits behind theorem-native trace closure rather than replacing it.

4.2 Execution protocol matrix

Domain	Wave	Evidence bar	Held-out policy	Replay command	Replay status	Closure verdict
Mathematics	WAVE_0_ANCHOR	HYBRID_ESCALATION	DETERMINISTIC_COUNTEREXAMPLE_AND_TRACE_REPLAY	run_claim_simulations_v1.py	PASS_REPLAYABLE	PASS
Physics	WAVE_1_PHYSICS	HYBRID_ESCALATION	LEAVE_ONE_BENCHMARK_FAMILY_OUT	run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS	PASS_REPLAYABLE	PASS

Domain	Wave	Evidence bar	Held-out policy	Replay command	Replay status	Closure verdict
Chemistry	WAVE_2_CHEMISTRY	HYBRID_ESCALATION	HOLD_OUT_ONE_COMPOUND_OR_REACTION_CLASS	run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY	PASS_REPLAYABLE	PASS
Biology	WAVE_3_BIOLOGY	HYBRID_ESCALATION	RESERVE_ONE_DATASET_FAMILY_PER_SIGNATURE_CLASS	run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY	PASS_REPLAYABLE	PASS
Systems / Civilizational projection	WAVE_4_SYSTEMS	HYBRID_ESCALATION	HELD_OUT_INTERVAL_REPLAY_WITH_TURNING_POINT_CHECKS	run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION	PASS_REPLAYABLE	PASS

4.3 Method ladder

Order	Method class	Entry rule	Anti-cycle condition
1	SOURCE_EXTRACTION	Use when a claim lacks an exact theorem target, exact dependency list, or exact source text.	May not be repeated unless a new source block, new theorem family, or newly scoped parent claim is introduced.
2	FRESH_DERIVATION_FROM_OPERATORS	Use when source extraction alone does not close the theorem-native route.	May not be repeated unless a new formal premise or a new decomposition of the claim space appears.
3	REPARAMETERIZATION_OR_CLAIM_DECOMPOSITION	Use when the claim is too wide, rubberized, or underspecified to admit a lawful parameter law.	May not be repeated unless a new observable family or a new admissible regime is declared before replay.
4	NEW_DATA_OR_MEASUREMENT_ROUTE	Use only after theorem target, parameter law, and held-out policy are already fixed.	May not be repeated unless the new route adds a previously unavailable observable family or a nonoverlapping held-out family.
5	SAME_CLAIM_CLASS_ADVERSARIAL_COMPARISON	Use when the claim survives internal closure steps and must now face simpler same-claim-class competitors.	May not be repeated unless a new comparator family or a new cost-normalized metric is introduced.

4.4 Per-domain escalation doctrine

Mathematics. Protocol OC13::EXECUTION::MATHEMATICS::ANCHOR_REPLAY runs under evidence bar HYBRID_ESCALATION and currently reports readiness PASS. Closure phase: PHASE_1_STABILIZE_CLOSED_CORE. Launch rule: MAINTAIN_ALWAYS_ON. Pinned route institutions: Clay Mathematics Institute, American Institute of Mathematics. Held-out policy: DETERMINISTIC_COUNTEREXAMPLE_AND_TRACE_REPLAY, with minimum cases 31. Quantitative gate: minimum cases 31; strictly terminalized cases at least 31; counterexample survivors capped at 0; replay divergences capped at 0. Replay harness: logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py. Institute-run trigger: ONLY_IF_DETERMINISTIC_REPLAY_BREAKS. Measurement wave: NONE. Open gaps: none. Escalation plan: Expand the exact-question corpus rather than broadening outward claims when a replay gap appears.

Physics. Protocol OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT runs under evidence bar HYBRID_ESCALATION and currently reports readiness PASS. Closure phase: PHASE_3_PHYSICS_CLOSURE. Launch rule: START_IMMEDIATELY_AFTER_PHASE_1. Pinned route institutions: NIST, U.S. Department of Energy Office of Science. Held-out policy: LEAVE_ONE_BENCHMARK_FAMILY_OUT, with minimum cases 30. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma; maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; critical residual threshold 2.5 sigma; severe residual threshold 1.5 sigma. Replay harness: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS. Institute-run trigger: OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_RESIDUAL_BREACH_PERSISTS. Measurement wave: INSTITUTE_RUN::PHYSICS::WAVE_1A. Open gaps: none. Escalation plan: If official/open data do not cover the required observable family, queue institute-run measurement or observational partner acquisition.

Chemistry. Protocol OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY_SPECTRA_KINETICS runs under evidence bar HYBRID_ESCALATION and currently reports readiness PASS. Closure phase: PHASE_4_CHEMISTRY_CLOSURE. Launch rule: START_AFTER_PHYSICS_THEOREM_PACKET_LOCKS. Pinned route institutions: NIST, U.S. Department of Energy Office of Science. Held-out policy: HOLD_OUT_ONE_COMPOUND_OR_REACTION_CLASS, with minimum cases 30. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma; maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; critical residual threshold 2.5 sigma; severe residual threshold 1.5 sigma. Replay harness: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY. Institute-run trigger: OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_RESIDUAL_BREACH_PERSISTS. Measurement wave: INSTITUTE_RUN::CHEMISTRY::WAVE_2A. Open gaps: none. Escalation plan: When official/open reference data are insufficient, queue targeted laboratory or partner-measurement acquisition.

Biology. Protocol OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPONSE_SIGNATURES runs under evidence bar HYBRID_ESCALATION and currently reports readiness PASS. Closure phase: PHASE_5_BIOLOGY_CLOSURE. Launch rule: START_AFTER_CHEMISTRY_THEOREM_PACKET_LOCKS. Pinned route institutions: NCBI GEO. Held-out policy: RESERVE_ONE_DATASET_FAMILY_PER_SIGNATURE_CLASS, with minimum cases 30. Quantitative gate: minimum cases 30; coverage ratio 1.0; mean absolute normalized error at or below 1.0 sigma; 95th-percentile normalized error at or below 2.5 sigma; maximum normalized error at or below 5.0 sigma; tail breaches capped at 0; expected calibration error at or below 0.05. Replay harness: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY. Institute-run trigger: OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_CASES_LT_30_OR_SIGNATURE_CLASS_UNDERCONSTRAINED. Measurement wave: INSTITUTE_RUN::BIOLOGY::WAVE_3A. Open gaps: none. Escalation plan: If open repositories cannot close the lane honestly, promote an institute-run observation or assay plan instead of overclaiming.

Systems / Civilizational projection. Protocol OC13::EXECUTION::SYSTEMS::MACRO_TRAJECTORIES_AND_REGIME_SHIFTS runs under evidence bar HYBRID_ESCALATION and currently reports readiness PASS. Closure phase: PHASE_6_SYSTEMS_CLOSURE. Launch rule: ALLOW_PARALLEL_DATA_PREP_AFTER_CHEMISTRY_BUT_FINAL_PROMOTION_AFTER_BIOLOGY_FORMAL_ROUTE. Pinned route institutions: World Bank, U.S. Department of Energy Office of Science. Held-out policy: HELD_OUT_INTERVAL_REPLAY_WITH_TURNING_POINT_CHECKS, with minimum cases 30. Quantitative gate: minimum cases 30; coverage ratio 1.0; tail breaches capped at 0; Brier score at or below 0.08; expected calibration error at or below 0.05; critical-failure F1 at least 0.87; critical-failure precision at least 0.85; critical-failure recall at least 0.9. Replay harness: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION. Institute-run trigger: OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_INTERVAL_COUNT_LT_30_OR_TURNING_POINT_BREACH_PERSISTS. Measurement wave: INSTITUTE_RUN::SYSTEMS::WAVE_4A. Open gaps: none. Escalation plan: If open macro-process series are insufficient, queue institute-run observation design or partner data acquisition.

Transition to proof machinery. Once the empirical bar is fixed, the next chapter closes the triad by stating the proof machinery, revision law, competition discipline, and compression rule that keep the outward manuscript synchronized to the science verdict.

5 Proof Machinery, Revision Law, Competition, and Compression

This chapter closes the Core 1.3 triad of operational discipline, empirical execution, and proof machinery. It fixes how proof obligations, revision law, comparator discipline, and compression rules are kept aligned with the canonical science state. It does not replace the theorem roadmap or the empirical protocols; it tells the reader how the release stays lawful when claims are challenged, revised, compared, or compressed for outward use. Comparison summaries, compression statements, and outward publication surfaces are projection-only editorial layers. They cannot alter the `CORE_1_3_SCIENCE_ONLY` truth scope, source-owned science corpus, closure verdicts, theorem refs, falsifier refs, formulas, or numerical rows.

This chapter is projected from the canonical science SPOT `OC_CORE_1_3_SCIENCE_SPOT` at `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json` so that the proof machinery and the science verdict stay synchronized.

The proof machinery is projected from one scientific authority. This keeps foundational consistency, empirical bridge status, revision law, and competition pressure under one coherent scientific verdict instead of across contradictory summary layers.

5.1 Consistency and promotion law

The foundational dossier tracks `13` K-level doctrine rows and sets the overall science verdict to `PASS`. The promotion law keeps two gates distinct. A derived claim is admissible only when it is source-bound, proof-bound, and dependency-exact. An empirical claim is promoted only under the full source-owned empirical gate: a domain packet, theorem-native trace, numerical parameter law, observable binding, pinned data route, locked held-out replay, comparator audit, and explicit falsifier must all be present and must survive the declared thresholds. This sentence is a compressed gate summary; the theorem roadmap records the item-by-item checklist. The hostile-review backlog is now fully locked, so no open dossier remains capable of blocking the promoted global PASS. Closure work is now executed through generated dossier packages and a first-class Phase 1 closed-core dossier rather than through drifting editorial reminders.

5.2 Revision law

The declared revision mutability remains `DERIVED_FIRST`, while the root revision status remains `LOCKED_PENDING_ACCUMULATED_CROSS_DOMAIN_FAILURE`.

5.3 Proof obligations

The load-bearing dependency atlas now tracks `6` explicit spine edges and `6` proof-obligation sheets. Together they replace vague prose summaries with explicit admissibility burdens, forbidden shortcuts, excluded dependencies, and falsifier conditions.

5.4 Minimality and irreducibility

The strongest honest meta-result remains `STRONGEST_IMPOSSIBILITY_BOUNDARY`, and K11/K12 now stand under a locked irreducibility result beyond K10.

5.5 Alternative-model competition

The competition policy remains `BOTH`: same-claim-class competition and domain-baseline competition are both in force. The matrix currently tracks `5` comparison rows, of which `4` are pass-eligible under the present SPOT.

5.6 Unified-science atlas

The unified atlas status is `PASS`. Shared operators are now globally locked across closed domains and no surviving cross-domain contradiction remains in the falsifier matrix.

5.7 Compression as a bounded metric

The release treats compression as `BOUNDED_METRIC` rather than as a central theorem claim. Coverage per rooted theorem-support statement remains `0.294118`, while coverage per active numerical packet remains `1.0`.

Transition to synthesis. With operational discipline, empirical execution, and proof machinery all fixed under one authority, the manuscript can now move to synthesis and then to the reader-facing practical question of what the framework presently enables.

A Empirical Validation Matrix and Replay Ledger

This appendix is projected from the canonical science SPOT so that bridge evidence and promotion verdicts remain visibly distinct.

This appendix is projected from the canonical SPOT. It reports both bridge evidence and promotion verdicts without letting one masquerade as the other.

Domain	Class	Trace	Evidence	Verdict	Blocking ids	Notes
Mathematics	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: Clay Mathematics Institute, American Institute of Mathematics; Replay: `logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py`; Metrics: Campaign `PASS_31_OF_31_TERMINALIZED`; 31 terminalized cases; Next action: `MAINTAIN_REPLAY_DISCIPLINE`
Physics	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: NIST, U.S. Department of Energy Office of Science; Replay: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS`; Metrics: 30 covered / 20 held-out cases; Mean abs. normalized error 0.188136 sigma; P95 normalized error 0.64778 sigma; Maximum normalized error 1.48 sigma; Tail breaches 0; Next action: `MAINTENANCE_ONLY`
Chemistry	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: NIST, U.S. Department of Energy Office of Science; Replay: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY`; Metrics: 30 covered / 19 held-out cases; Mean abs. normalized error 0.533285 sigma; P95 normalized error 1.0 sigma; Maximum normalized error 1.597772 sigma; Tail breaches 0; Next action: `MAINTENANCE_ONLY`
Biology	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: NCBI GEO; Replay: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY`; Metrics: 30 covered / 16 held-out cases; Mean abs. normalized error 0.444222 sigma; P95 normalized error 1.030033 sigma; Maximum normalized error 1.438582 sigma; Tail breaches 0; Next action: `MAINTENANCE_ONLY`
Systems / Civilizational projection	THEOREM_NATIVE	TRACE_COMPLETE	EVIDENCE_COMPLETE	PASS	none	Sources: World Bank, U.S. Department of Energy Office of Science; Replay: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION`; Metrics: 30 covered / 18 held-out cases; Mean abs. normalized error 0.56097 sigma; P95 normalized error 1.924704 sigma; Maximum normalized error 2.25395 sigma; Tail breaches 0; Next action: `MAINTENANCE_ONLY`

B Institute-Run Measurement and Observation Program

This appendix records the stand-by measurement waves that would become mandatory only if pinned official or open sources stopped closing the declared empirical lanes honestly. At the present release boundary, all four empirical domains remain on official- or open-data routes; the institute-run waves are retained as controlled fallback mechanisms rather than as active evidence channels.

Domain	Stand-by wave	Escalation trigger	Current action
Physics	`INSTITUTE_RUN::PHYSICS::WAVE_1A`	Open-data coverage falls below full coverage, held-out coverage drops below 30 cases, or residual breaches persist after replay maintenance.	Maintain replay discipline
Chemistry	`INSTITUTE_RUN::CHEMISTRY::WAVE_2A`	Open-data coverage falls below full coverage, held-out coverage drops below 30 cases, or residual breaches persist after replay maintenance.	Maintain replay discipline
Biology	`INSTITUTE_RUN::BIOLOGY::WAVE_3A`	Open-data coverage falls below full coverage, held-out coverage drops below 30 cases, or the bounded signature class remains underconstrained.	Maintain replay discipline
Systems / Civilizational projection	`INSTITUTE_RUN::SYSTEMS::WAVE_4A`	Open-data coverage falls below full coverage, held-out interval coverage drops below 30 intervals, or turning-point breaches persist.	Maintain replay discipline

Interpretive rule. These waves are not alternate proof routes. They are reserve observation programs that become lawful only after the official or open route fails to cover the declared observable family. Until such a trigger is reached, the current release remains committed to the pinned replay discipline already documented in the empirical protocol chapters and benchmark appendices.

C Pinned Benchmark Dataset Manifest

This appendix records the pinned benchmark families, held-out policies, quantitative gates, and falsifiers that define lawful empirical promotion in the current release. The point is not to restate raw registry fields one by one, but to show which dataset family carries which evidential burden and what would count as a genuine empirical failure.

Mathematics. Protocol `OC13::EXECUTION::MATHEMATICS::ANCHOR_REPLAY` runs under the hybrid-escalation evidence bar, but in practice this lane is deterministic rather than measurement-driven. The held-out policy is `DETERMINISTIC_COUNTEREXAMPLE_AND_TRACE_REPLAY`, with a minimum of 31 strict cases. Coverage, sigma thresholds, Brier score, and ECE are therefore not applicable. Promotion survives only if no replay divergence and no surviving counterexample appear. Replay command: `logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py`. No external dataset manifest is required because the mathematics anchor closes on strict replay and counterexample traces rather than on observational data.

Physics. Protocol `OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT` runs under hybrid escalation and binds theorem-to-observable maps to pinned official constants, spectral tables, and transport challenge corpora. The current caseset contains 30 pinned cases, 20 held-out cases, and declared predictions for all 30 benchmark rows. The quantitative gate requires full coverage, mean absolute normalized error at most 1.0 sigma, p95 at most 2.5 sigma, maximum error at most 5.0 sigma, and zero tail breaches. Promotion survives only if every benchmark family remains within tolerance; any residual breach or broken sign/order constraint falsifies the packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-id PHYSICS`.

Domain	Execution protocol	Held-out design	Quantitative gate
Mathematics	`OC13::EXECUTION::MATHEMATICS::ANCHOR_REPLAY`	Deterministic counterexample and trace replay; minimum 31 cases	No sigma gate; promotion fails on any divergence or surviving counterexample
Physics	`OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT`	Leave one benchmark family out; minimum 30 cases	Coverage 1.0; mean abs. error at most 1.0 sigma; p95 at most 2.5 sigma; max at most 5.0 sigma; tail breaches 0
Chemistry	`OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY_SPECTRA_KINETICS`	Hold out one compound or reaction class; minimum 30 cases	Coverage 1.0; mean abs. error at most 1.0 sigma; p95 at most 2.5 sigma; max at most 5.0 sigma; tail breaches 0
Biology	`OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPONSE_SIGNATURES`	Reserve one dataset family per signature class; minimum 30 cases	Coverage 1.0; mean abs. error at most 1.0 sigma; p95 at most 2.5 sigma; max at most 5.0 sigma; ECE at most 0.05; tail breaches 0
Systems / Civilizational projection	`OC13::EXECUTION::SYSTEMS::MACRO_TRAJECTORIES_AND_REGIME_SHIFTS`	Held-out interval replay with turning-point checks; minimum 30 cases	Coverage 1.0; Brier at most 0.08; ECE at most 0.05; tail breaches 0

- PHY_BENCH_001: fine_structure_constant_residual. Authority: NIST/CODATAconstants. Dataset: https://physics.nist.gov/cuu/Constants/index.html. Held-out policy: leave one family out for residual verification.
- PHY_BENCH_002: hydrogen_balmer_line_residual. Authority: NISTAtomicSpectraDatabase. Dataset: https://physics.nist.gov/PhysRefData/ASD/lines_form.html. Held-out policy: reserve transitions not used during packet fitting.
- PHY_BENCH_003: transport_scaling_residual. Authority: DOEOfficeofSciencechallengecorpora. Dataset: https://science.osti.gov/-/media/fes/pdf/workshop-reports/FES_Grand_Challenges_Report_final.pdf. Held-out policy: hold out one transport-regime family.

Chemistry. Protocol OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY_SPECTRA_KINETICS maps theorem packets to thermochemical, spectral, and kinetic observables against pinned reference tables. The current caseset contains 30 pinned cases, 19 held-out cases, and declared predictions for all 30 benchmark rows. The quantitative gate matches the physics lane: full coverage, mean absolute normalized error at most 1.0 sigma, p95 at most 2.5 sigma, maximum error at most 5.0 sigma, and zero tail breaches. A held-out thermochemical, spectral, or kinetic family outside tolerance falsifies the promoted chemistry packet. Replay command: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY.

- CHEM_BENCH_001: gas_phase_enthalpy_residual. Authority: NISTChemistryWebBook. Dataset: https://webbook.nist.gov/chemistry/. Held-out policy: hold out one compound family from calibration.
- CHEM_BENCH_002: spectral_transition_residual. Authority: NISTChemistryWebBook. Dataset: https://webbook.nist.gov/chemistry/. Held-out policy: hold out one spectral family.
- CHEM_BENCH_003: kinetic_or_equilibrium_residual. Authority: DOEBES/EFRCchallengecorpora. Dataset: https://science.osti.gov/bes/efrc/Research/Grand-Challenges. Held-out policy: hold out one reaction-class envelope.

Biology. Protocol OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPONSE_SIGNATURES chooses one bounded biological lane at a time and evaluates predicted state transitions or response signatures against open assay data. The current caseset contains 30 pinned cases, 16 held-out cases, and declared predictions for all 30 benchmark rows. The quantitative gate requires full coverage, mean absolute normalized error at most 1.0 sigma, p95 at most 2.5 sigma, maximum error at most 5.0 sigma, ECE at most 0.05, and zero tail breaches. Promotion survives only if the bounded signature family reproduces the held-out dataset family without widening the claim boundary. Replay command: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY.

- BIO_BENCH_001: expression_signature_residual. Authority: NCBIGEO. Dataset: https://www.ncbi.nlm.nih.gov/geo/. Held-out policy: reserve one time-series family for held-out replay.
- BIO_BENCH_002: state_transition_order_error. Authority: NCBIGEO. Dataset: https://www.ncbi.nlm.nih.gov/geo/. Held-out policy: reserve one cell-state transition family.
- BIO_BENCH_003: response_onset_residual. Authority: official open biological assay repositories. Dataset: https://www.ncbi.nlm.nih.gov/geo/. Held-out policy: reserve one response-signature family.

Systems / Civilizational projection. Protocol OC13::EXECUTION::SYSTEMS::MACRO_TRAJECTORIES_AND_REGIME_SHIFTS evaluates K8 projection packets against pinned macro-process series with held-out interval tests. The current caseset contains 30 pinned cases, 18 held-out cases, and declared predictions for all 30 benchmark rows. The quantitative gate requires full coverage, Brier score at most 0.08, ECE at most 0.05, and zero tail breaches. Promotion survives only if the declared trajectory and regime-change observables remain inside tolerance on held-out intervals; a missed turning point or out-of-band trajectory falsifies the packet. Replay command: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION.

- SYS_BENCH_001: population_trajectory_residual. Authority: WorldBankIndicatorsAPI. Dataset: https://api.worldbank.org/v2/en/indicator/SP.POP.TOTL?format=json. Held-out policy: leave out one interval family.
- SYS_BENCH_002: gdp_growth_turning_point_error. Authority: WorldBankIndicatorsAPI. Dataset: https://api.worldbank.org/v2/en/indicator/NY.GDP.MKTP.KD.ZG?format=json. Held-out policy: reserve one held-out macro interval.
- SYS_BENCH_003: process_system_regime_score. Authority: WorldBankIndicatorsAPI. Dataset: https://api.worldbank.org/v2/en/indicator/EG.USE.PCAP.KG.OE?format=json. Held-out policy: reserve one resource-intensity curve family.

D Domain Replay Reports

This appendix collects the replay packets that the empirical closure program actually executes. The summary table gives the operational overview. The domain notes that follow explain what each replay lane is meant to verify and what would count as a real failure rather than as cosmetic noise.

Domain	Replay state	Caseset summary	Key metric snapshot	Wave
Mathematics	PASS_REPLAYABLE	Deterministic strict replay; minimum 31 cases	No divergence; no surviving counterexample	not required
Physics	PASS_REPLAYABLE	30 covered / 20 held-out cases	Mean abs. 0.188136 sigma; p95 0.64778 sigma; max 1.48 sigma; tail breaches 0	INSTITUTE_RUN::PHYSICS::WAVE_1A
Chemistry	PASS_REPLAYABLE	30 covered / 19 held-out cases	Mean abs. 0.533285 sigma; p95 1.0 sigma; max 1.597772 sigma; tail breaches 0	INSTITUTE_RUN::CHEMISTRY::WAVE_2A
Biology	PASS_REPLAYABLE	30 covered / 16 held-out cases	Mean abs. 0.444222 sigma; p95 1.030033 sigma; max 1.438582 sigma; tail breaches 0	INSTITUTE_RUN::BIOLOGY::WAVE_3A
Systems / Civilizational projection	PASS_REPLAYABLE	30 covered / 18 held-out cases	Mean abs. 0.56097 sigma; p95 1.924704 sigma; max 2.25395 sigma; tail breaches 0	INSTITUTE_RUN::SYSTEMS::WAVE_4A

Mathematics. The mathematics anchor is a deterministic replay lane rather than an empirical benchmark family. It currently sits at verdict `PASS` with replay state `PASS_REPLAYABLE`. No parameterized cases are required because the lane tests strict closure ledgers and counterexample traces directly. Promotion fails on the first replay divergence or on the first surviving counterexample. Replay command: `logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py`. The lawful next action remains `MAINTAIN_REPLAY_DISCIPLINE`.

Physics. The promoted physics packet is `OC13::PHYSICS::FOUNDATIONAL_CONSTANTS_AND_SPECTRA`, executed through `OC13::EXECUTION::PHYSICS::CONSTANTS_SPECTRA_TRANSPORT`. Held-out residuals must remain within the declared tolerance band for every promoted benchmark family. Any family outside tolerance, or any broken sign/order constraint, falsifies the packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS`. The current official replay basis is `PARTIAL_OFFICIAL_PRIMARY_DATA`, with 30 covered cases, 20 held-out cases, mean absolute normalized error 0.188136 sigma, p95 0.64778 sigma, maximum 1.48 sigma, and zero tail breaches.

- `PHY_BENCH_001`: `fine_structure_constant_residual` from https://physics.nist.gov/cuu/Constants/index.html; held-out policy: leave one family out for residual verification.
- `PHY_BENCH_002`: `hydrogen_balmer_line_residual` from https://physics.nist.gov/PhysRefData/ASD/lines_form.html; held-out policy: reserve transitions not used during packet fitting.
- `PHY_BENCH_003`: `transport_scaling_residual` from https://science.osti.gov/-/media/fes/pdf/workshop-reports/FES_Grand_Challenges_Report_final.pdf; held-out policy: hold out one transport-regime family.

Chemistry. The promoted chemistry packet is `OC13::CHEMISTRY::THERMOCHEMISTRY_AND_SPECTRA`, executed through `OC13::EXECUTION::CHEMISTRY::THERMOCHEMISTRY_SPECTRA_KINETICS`. The declared thermochemical, spectral, and kinetic benchmark families must remain within tolerance on held-out compounds or reaction classes. A held-out family outside tolerance falsifies the packet. Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY`. The official replay basis remains `PARTIAL_OFFICIAL_PRIMARY_DATA`, with 30 covered cases, 19 held-out cases, mean absolute normalized error 0.533285 sigma, p95 1.0 sigma, maximum 1.597772 sigma, and zero tail breaches.

- CHEM_BENCH_001: gas_phase_enthalpy_residual from https://webbook.nist.gov/chemistry/; held-out policy: hold out one compound family from calibration.
- CHEM_BENCH_002: spectral_transition_residual from https://webbook.nist.gov/chemistry/; held-out policy: hold out one spectral family.
- CHEM_BENCH_003: kinetic_or_equilibrium_residual from https://science.osti.gov/bes/efrc/Research/Grand-Challenges; held-out policy: hold out one reaction-class envelope.

Biology. The promoted biology packet is OC13::BIOLOGY::STATE_TRANSITION_AND_RESPONSE_SIGNATURES, executed through OC13::EXECUTION::BIOLOGY::STATE_TRANSITIONS_AND_RESPONSE_SIGNATURES. The bounded measurable signatures must reproduce the held-out dataset families without widening the claim boundary. If the signature family fails on the held-out datasets, the biological lane loses promotion for the affected packet. Replay command: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY. The official replay basis is OFFICIAL_OPEN_PRIMARY_DATA, with 30 covered cases, 16 held-out cases, mean absolute normalized error 0.444222 sigma, p95 1.030033 sigma, maximum 1.438582 sigma, and zero tail breaches.

- BIO_BENCH_001: expression_signature_residual from https://www.ncbi.nlm.nih.gov/geo/; held-out policy: reserve one time-series family for held-out replay.
- BIO_BENCH_002: state_transition_order_error from https://www.ncbi.nlm.nih.gov/geo/; held-out policy: reserve one cell-state transition family.
- BIO_BENCH_003: response_onset_residual from https://www.ncbi.nlm.nih.gov/geo/; held-out policy: reserve one response-signature family.

Systems / Civilizational projection. The promoted systems packet is OC13::SYSTEMS::K8_MACRO_PROCESS_AND_REGIME_SHIFT, executed through OC13::EXECUTION::SYSTEMS::MACRO_TRAJECTORIES_AND_REGIME_SHIFTS. The declared trajectory and regime-change observables must remain within tolerance on held-out intervals; a missed turning point or an out-of-band trajectory falsifies the packet. Replay command: logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION. The official replay basis is OFFICIAL_OPEN_PRIMARY_DATA, with 30 covered cases, 18 held-out cases, mean absolute normalized error 0.56097 sigma, p95 1.924704 sigma, maximum 2.25395 sigma, and zero tail breaches.

- SYS_BENCH_001: population_trajectory_residual from https://api.worldbank.org/v2/en/indicator/SP.POP.TOTL?format=json; held-out policy: leave out one interval family.
- SYS_BENCH_002: gdp_growth_turning_point_error from https://api.worldbank.org/v2/en/indicator/NY.GDP.MKTP.KD.ZG?format=json; held-out policy: reserve one held-out macro interval.
- SYS_BENCH_003: process_system_regime_score from https://api.worldbank.org/v2/en/indicator/EG.USE.PCAP.KG.OE?format=json; held-out policy: reserve one resource-intensity curve family.

E Domain Hard-Closure Command Board

This appendix turns the empirical closure backlog into an execution board that a reviewer can actually read. The summary table states which domains remain in maintenance mode. The domain notes make explicit which command constitutes the lawful next step, which acceptance rule that command is meant to protect, and which reserve measurement wave remains on stand-by.
The current board exposes five tracked lanes. All five are validated and sit in maintenance mode; none presently requires an active institute-run fallback wave.

Mathematics. Validation remains PASS, and the lane is executable now because the command path is purely deterministic. Caseset readiness is 0 parameterized out of 0 cases with a minimum requirement of 31, while coverage and parameterization remain NOT_REQUIRED. Promotion survives only if no replay divergence and no surviving counterexample appear. The lawful command is logion/k7/spe/orchestrator/science/run_claim_simulations_v1.py. The required prerequisites remain the pinned source route, the benchmark manifest, the numerical packet specification, the replay harness, and an explicit falsifier. Blocking ids: none. No institute-run wave is required.

Physics. Validation remains PASS, and the lane stays in maintenance rather than in expansion mode. Caseset readiness is 30 parameterized out of 30 cases, with coverage PASS_CASESET_MINIMUM_REACHED and parameterization PREDICTIONS_DECLARED_FOR_ALL_CASES. The acceptance criterion requires held-out residuals to stay within the declared tolerance band for every promoted benchmark family; any family outside tolerance or with broken sign/order constraints falsifies the packet. The lawful command is logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS. The reserve wave INSTITUTE_RUN::PHYSICS::WAVE_1A remains on stand-by, not active.

Domain	Validation	Phase	Lawful command	Next action
Mathematics	PASS	MAINTENANCE	run_claim_simulations_v1.py	MAINTAIN_REPLAY_DISCIPLINE
Physics	PASS	MAINTENANCE	run_oc_core_domain_hard_closure_replay_v1.py--domain-idPHYSICS	MAINTAIN_REPLAY_DISCIPLINE
Chemistry	PASS	MAINTENANCE	run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY	MAINTAIN_REPLAY_DISCIPLINE
Biology	PASS	MAINTENANCE	run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY	MAINTAIN_REPLAY_DISCIPLINE
Systems / Civilizational projection	PASS	MAINTENANCE	run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION	MAINTAIN_REPLAY_DISCIPLINE

Chemistry. Validation remains PASS, and the chemistry lane also sits in maintenance mode. Caseset readiness is 30 parameterized out of 30 cases, with coverage PASS_CASESET_MINIMUM_REACHED and parameterization PREDICTIONS_DECLARED_FOR_ALL_CASES. The acceptance criterion requires the declared chemistry benchmark families to remain within tolerance on held-out compounds or reactions; a held-out thermochemical, spectral, or kinetic family outside tolerance falsifies the packet. The lawful command is logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idCHEMISTRY. The reserve wave INSTITUTE_RUN::CHEMISTRY::WAVE_2A remains on stand-by.

Biology. Validation remains PASS, and the biological lane is executable now only in maintenance mode. Caseset readiness is 30 parameterized out of 30 cases, with coverage PASS_CASESET_MINIMUM_REACHED and parameterization PREDICTIONS_DECLARED_FOR_ALL_CASES. The acceptance criterion requires the promoted biological packet to reproduce the declared measurable signatures on held-out datasets without widening the claim boundary. If the bounded signature family fails on the held-out datasets, the lane loses promotion. The lawful command is logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idBIOLOGY. The reserve wave INSTITUTE_RUN::BIOLOGY::WAVE_3A remains on stand-by.

Systems / Civilizational projection. Validation remains PASS, and the systems lane is likewise executable only in maintenance mode. Caseset readiness is 30 parameterized out of 30 cases, with coverage PASS_CASESET_MINIMUM_REACHED and parameterization PREDICTIONS_DECLARED_FOR_ALL_CASES. The acceptance criterion requires the declared trajectory and regime-change observables to remain within tolerance on held-out intervals; a missed turning point or out-of-band held-out trajectory falsifies the packet. The lawful command is logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py--domain-idSYSTEMS_CIVILIZATIONAL_PROJECTION. The reserve wave INSTITUTE_RUN::SYSTEMS::WAVE_4A remains on stand-by.

F Proof Machinery Appendix

The source-owned science corpus is the authoring and witness locus. The canonical SPOT surface releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json is the synchronised scientific projection authority used for audit and navigation. References to both below are provenance links, not a division of authority between two independent truth layers. Source stamp: corpus path releases/oc_core_1_3/editorial/science_sources/; governing surface OC_CORE_1_3_SCIENCE_SPOT_latest.json; snapshot fields metadata.repo_sha and metadata.ts_utc. The generated material below is a compact proof-machinery atlas, not a replacement for the full proof-obligation sheets. It opens with release revision law and then projects the load-bearing proof obligations, hostile- review backlog, closure priority, ablation, competition, and compression surfaces. Premises, forbidden shortcuts, excluded dependencies, and falsifier conditions are authoritative in releases/oc_c ore_1_3/editorial/OC_CORE_1_3_PROOF_OBLIGATION_REGISTRY_latest.json; auxiliary rows remain navigation summaries derived from the canonical SPOT and its release surfaces rather than independent proof authority.

F.1 Revision law authority

Priority	Layer	Revision rule	Conflict trigger	Current status
1	EMPIRICAL_BRIDGES_AND_REPLAY_BINDINGS	REVISE_EMPIRICAL_BRIDGES_FIRST	OFFICIAL_OR_INSTITUTE_RUN_REPLAY_BREACH	STABLE
2	PARAMETERIZATION_AND_NUMERICAL_PACKETS	REVISE_PARAMETERIZATIONS_BEFORE_THEOREM_SCOPE	RESIDUAL_OR_CALIBRATION_BREACH_PERSISTS_AFTER_BRIDGE_REPAIR	STABLE
3	STANDALONE_DOMAIN_PACKETS_AND_DERIVED_THEOREMS	REVISE_STANDALONE_DOMAIN_PACKETS_BEFORE_ROOT_KERNEL	MULTI_BENCHMARK_FAILURE_PERSISTS_AFTER_PARAMETERIZATION_REPAIR	REAUDIT_REQUIRED_FOR_BRIDGE_ONLY_DOMAINS
4	ROOT_KERNEL	ROOT_KERNEL_REVISION_ALLOWED_ONLY_AFTER_ACCUMULATED_CROSS_DOMAIN_FAILURE	ACCUMULATED_CROSS_DOMAIN_FAILURE_EXCEEDS_DECLARED_THRESHOLD	LOCKED_PENDING_ACCUMULATED_CROSS_DOMAIN_FAILURE

F.2 Load-bearing proof obligations

Obligation id	Statement	Excluded dependencies	Falsifier condition
OBLIGATION::SOURCE_THEOREM_3	Show that Axiom 3.2 plus Definitions 12.1, 12.3, and 12.4 lawfully imply Source Theorem 3.	Excluded lift theorem family, Journal-core-only compression	If the implication requires a hidden excluded dependency or the admissibility premises do not suffice, Source Theorem 3 must be narrowed or withdrawn.
OBLIGATION::LEMMA_1	Show that Source Theorem 3 closes Lemma 1 without widening the claim boundary.	Interpretive-only support, Compression-only support	If Lemma 1 requires support not traceable to Source Theorem 3, the spine is broken and the positive body must be rebuilt.
OBLIGATION::SOURCE_COROLLARY_3_2	Show that Axiom 3.3 plus Definition 12.6 lawfully imply Source Corollary 3.2 and preserve irreversibility.	Identity restoration after lawful death, Metaphorical rebirth language	If the same continuum can return numerically identical after lawful death without violating Axiom 3.3, the irreversibility corollary must be revised or withdrawn.
OBLIGATION::LEMMA_3	Show that Source Corollary 3.2 closes Lemma 3 under the declared residue and rebirth boundary.	Persistence-by-analogy, Undeclared rebirth equivalence	If residue preserves full live identity instead of structural aftermath only, the classification boundary must be rewritten.
OBLIGATION::LEMMA_2	Show that Lemma 1 plus Definition 12.5 imply Lemma 2 without hidden extensions.	EXTENSION branches, BRIDGE_ONLY packets	If Lemma 2 requires an excluded branch or bridge-only evidence as proof support, the theorem route is illegitimate.
OBLIGATION::THEOREM_A	Show that Lemma 2 is sufficient for Theorem A and that the journal-core extraction does not exceed the master theorem scope.	Journal-core-only strengthening, Excluded lift-heavy theorem family	If the journal core requires support not present in the master route, the article must be narrowed or the master must be repaired first.

F.3 Hostile-review backlog

Review id	Challenge	Status	Resolution state	Exit criterion
HOSTILE::LAWFUL_COLLAPSE	Collapse must be tied to the loss of admissible realization rather than to loose metaphor.	PASS	PASS_LOCKED	A formal dossier links collapse claims to admissible-state loss, threshold logic, and source witnesses without prose leakage.
HOSTILE::IRREVERSIBILITY	Irreversibility under Axiom 3.3 must forbid numerical identity restoration after lawful death.	PASS	PASS_LOCKED	The irreversibility corollary is formalized as a dossier with explicit counterexample conditions and no identity loophole.
HOSTILE::RESIDUE_REBIRTH_BOUNDARY	Residue and rebirth must remain controlled classifications rather than narrative labels.	PASS	PASS_LOCKED	Residue, rebirth, and persistence are separated by theorem-backed boundary conditions with declared falsifiers.
HOSTILE::EXCLUDED_LIFT_FIREWALL	No excluded lift-heavy theorem family may be smuggled into the positive bounded argument.	PASS	PASS_LOCKED	An exclusion ledger proves that forbidden theorem families and forbidden branches do not enter the core proof path.
HOSTILE::JOURNAL_CORE_ASYMMETRY	The journal core must remain a lawful extraction from the master and must never enlarge the claim.	PASS	PASS_LOCKED	A journal-core extraction dossier proves that every public theorem claim is traceably derived from the master route.
HOSTILE::K11_K12_IRREDUCIBILITY	K11 and K12 must be proved necessary beyond K10 or explicitly demoted with the dependency graph rebuilt.	PASS	PROVED_NECESSARY_BEYOND_K10	Either an irreducibility dossier proves that K11 and K12 carry nonreducible scientific work, or the upper levels are demoted and all dependent routes are rebuilt.

F.4 Closure priority board

Domain	Formal order	Cost order	Gain per cost	Phase	Parallelism note
Mathematics	0	0	116.666667	PHASE_1_STABILIZE_CLOSED_CORE	Runs continuously as the formal anchor for all later domains.
Physics	1	1	64.8	PHASE_3_PHYSICS_CLOSURE	Primary first closure lane; no later domain should outrun it.
Chemistry	2	2	32.0	PHASE_4_CHEMISTRY_CLOSURE	Next formal lane after physics; some data preparation may run earlier, but promotion waits for the physics shell to lock.
Biology	3	4	17.818182	PHASE_5_BIOLOGY_CLOSURE	Formal order precedes systems, but the high measurement cost makes it a slower lane.
Systems / Civilizational projection	4	3	16.615385	PHASE_6_SYSTEMS_CLOSURE	Cheaper empirical prep than biology is allowed, but final unification waits for the biological formal route.

F.5 Minimality and irreducibility ablation ledger

Component	Kind	Ablation status	Removal effect
ROOT_THEOREM_STACK	KERNEL_COMPONENT	NECESSARY_UNDER_CURRENT_PROOF_STACK	Collapses the promoted contradiction-lift-or-collapse doctrine.
K_LEVEL_PARTITION	K_LEVEL_BOUNDARY	FRONTIER_PENDING_STRICT_IRREDUCIBILITY_PROOF	Would blur the explicit K-level doctrine and weaken the current boundary ledger.
EMPIRICAL_ANCHORING_DISCIPLINE	PROMOTION_GUARD	NECESSARY_FOR_NON_RUBBERIZING_PROMOTION	Would allow positive outward science without observables, data routes, and replayable falsifiers.
STRICT_UNIQUENESS_OR_MINIMALITY	META_THEOREM	EXPLICIT_FRONTIER_RESIDUE_LOCALIZED	The strongest current result remains an impossibility boundary rather than a strict uniqueness proof.

F.6 Alternative-model competition matrix

Competition id	Class	Evaluation status	Comparison scope
SAME_CLAIM_CLASS::META_MODEL	SAME_CLAIM_CLASS_METAMODEL	PENDING_COMPLEXITY_PENALIZED_SAME_CLAIM_CLASS_COMPARISON	Whole-kernel explanatory and predictive claim class
DOMAIN_BASELINE::PHYSICS	DOMAIN_BASELINE	DOMAIN_BASELINE_COMPARISON_PASS_ELIGIBLE	Bounded theorem-native physics packet is trace-closed, replay-closed, and comparator-clean on the pinned official held-out route.
DOMAIN_BASELINE::CHEMISTRY	DOMAIN_BASELINE	DOMAIN_BASELINE_COMPARISON_PASS_ELIGIBLE	Bounded theorem-native chemistry packet is trace-closed, replay-closed, and comparator-clean on the pinned thermochemical, spectral, and kinetic held-out route.
DOMAIN_BASELINE::BIOLOGY	DOMAIN_BASELINE	DOMAIN_BASELINE_COMPARISON_PASS_ELIGIBLE	Bounded theorem-native biology packet is trace-closed, replay-closed, and comparator-clean across the pinned multi-dataset held-out biological lanes.
DOMAIN_BASELINE::SYSTEMS_CIVILIZATIONAL_PROJECTION	DOMAIN_BASELINE	DOMAIN_BASELINE_COMPARISON_PASS_ELIGIBLE	Bounded theorem-native systems packet is trace-closed, replay-closed, and comparator-clean on the pinned macro-trajectory and regime-shift held-out route.

F.7 Bounded compression benchmark

Axis	Complexity units	Performance value	Interpretation
THEOREM_SUPPORT_DESCRIPTION_LENGTH	17	0.294118	How much validated domain coverage is currently carried per rooted theorem-support statement.
EMPIRICAL_PACKET_COMPLEXITY_VS_RESIDUAL_PERFORMANCE	5	1.0	How much validated coverage is carried per active numerical packet under the present replay bar.

G OC 1.3.1 Simulation and Data Appendix

- lane_id=LANE::MATHEMATICS; domain=mathematics; strengthening_status=STRENGTHENED_NUMERICALLY; primary_strengthening_mode=NUMERIC_PACKET
- lane_id=LANE::PHYSICS; domain=physics; strengthening_status=STRENGTHENED_NUMERICALLY; primary_strengthening_mode=NUMERIC_PACKET

- lane_id=LANE::CHEMISTRY; domain=chemistry; strengthening_status=STRENGTHENED_NUMERICALLY; primary_strengthening_mode=NUMERIC_PACKET
- lane_id=LANE::K3_K4_ORIGINS; domain=k3_k4_origins_of_life; strengthening_status=STRENGTHENED_BY_SI primary_strengthening_mode=SIMULATION_CORPUS
- lane_id=LANE::BIOLOGY_K4_K5; domain=biology; strengthening_status=STRENGTHENED_BY_DATA_ROUTE primary_strengthening_mode=DATA_MANIFEST_BINDING
- lane_id=LANE::COGNITION_K6; domain=k6_cognition; strengthening_status=STRENGTHENED_FORMALLY; primary_strengthening_mode=FORMULA_AND_OPERATOR_PATCH
- lane_id=LANE::SOCIETY_K7; domain=k7_society; strengthening_status=STRENGTHENED_FORMALLY; primary_strengthening_mode=FORMULA_AND_OPERATOR_PATCH
- lane_id=LANE::SYSTEMS_K8; domain=k8_systems; strengthening_status=STRENGTHENED_BY_SIMULATION; primary_strengthening_mode=SIMULATION_CORPUS
- lane_id=LANE::KNOWLEDGE_K9; domain=k9_knowledge_systems; strengthening_status=STRENGTHENED_FO primary_strengthening_mode=FORMULA_AND_OPERATOR_PATCH
- lane_id=LANE::FORMAL_K10; domain=k10_formal_systems; strengthening_status=STRENGTHENED_FORMAL primary_strengthening_mode=FORMULA_AND_OPERATOR_PATCH
- lane_id=LANE::META_K11_K12; domain=k11_k12_meta_theory; strengthening_status=PARTIAL_ONLY; primary_strengthening_mode=FRONTIER_AND_SIMULATION_MIX

H OC 1.3.1 Science Delta Appendix

- delta_id=DELTA::WHITE_SPOT_CLOSURE; delta_class=scientific_closure; status_1_3_1=closure_companion_and action_bucket=must_be_added
- delta_id=DELTA::SIMULATION_CORPUS; delta_class=simulation_and_reproducibility; status_1_3_1=top_level_simulation_corpus; action_bucket=must_be_added
- delta_id=DELTA::DATA_MANIFEST_LAYER; delta_class=data_route_and_traceability; status_1_3_1=manifest_anc action_bucket=must_be_added
- delta_id=DELTA::MASTER_MONOGRAPH; delta_class=release_authority; status_1_3_1=oc_core_1_3_1_master_p action_bucket=must_be_added
- delta_id=DELTA::ARTICLE_FAMILY; delta_class=publication_surface; status_1_3_1=bounded_article_family_prese action_bucket=must_be_added
- delta_id=DELTA::RELEASE_AUTOMATION; delta_class=release_automation; status_1_3_1=contract_and_validato action_bucket=must_be_added
- delta_id=DELTA::NO_SEND_OUTBOUND; delta_class=outbound_readiness; status_1_3_1=release_specific_packag action_bucket=must_be_added
- delta_id=DELTA::PUBLIC_RELEASE_STAGE; delta_class=release_staging; status_1_3_1=oc_core_1_3_1_ready_n action_bucket=candidate_for_1_3_1
- delta_id=DELTA::K11_K12_EXTENSION; delta_class=high_cost_theory_extension; status_1_3_1=mapped_but_not action_bucket=must_remain_frontier
- delta_id=DELTA::PAID_DATA_ESCALATIONS; delta_class=data_escalation; status_1_3_1=still_optional; action_bucket=too_expensive_now